- **Title :** Create a small web-based application by selecting relevant system environments/platforms and programming languages. Narrate concise Test Plan consisting features to be tested and bug taxonomy.Narrate scripts in order to perform regression tests. Identify the bugs using Selenium WebDriver and IDEand generate test reports encompassing exploratory testing.

- **Problem Definition:** Perform Web testing and identify the bugs using Selenium WebDriver and IDE and generate test reports for (your **mini project name**)
- **Prepared By:**

  Name, Roll no., BE COMP

  PQR, BCxxx BE COMP

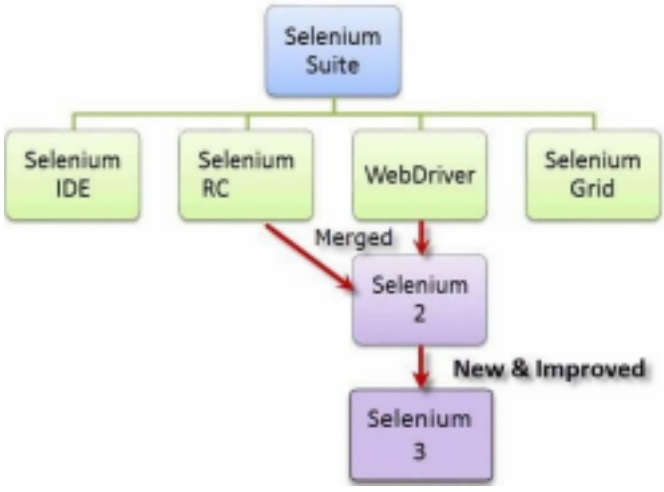
- **Prerequisite:** Knowledge of Core Java
- **Software Requirements:**
- **Hardware Requirement:**
- Theory :

Introduction of mini project

Testing Tool: Selenium

Selenium is a free (open-source) automated testing suite for web applications across different browsers and platforms. It is quite similar to HP Quick Test Pro (QTP now UFT) only that Selenium focuses on automating web-based applications. Testing done using the Selenium tool is usually referred as Selenium Testing. Selenium is not just a single tool but a suite of software, each catering to different testing needs of an organization. It has four components.

- Selenium Integrated Development Environment(IDE)
- Selenium Remote Control (RC)
- Web Driver
- Selenium Grid



Selenium IDE (Integrated Development Environment) is the simplest tool in the Selenium Suite. It is a Firefox add-on that creates tests very quickly through its record- and-playback functionality. This feature is similar to that of QTP. It is effortless to install and easy to learn. Because of its simplicity, Selenium IDE should only be used as a prototyping tool, not an overall solution for developing and maintaining complex test suites.

Though you will be able to use Selenium IDE without prior knowledge in programming, you should at least be familiar with HTML, JavaScript, and the DOM (Document Object Model) to utilize this tool to its full potential. Knowledge of JavaScript will be required when we get to the section about the Selenese command "run Script."

Selenium IDE supports auto-complete mode when creating tests. This feature serves two purposes: It helps the tester to enter commands more quickly.
It restricts the user from entering invalid commands.

The way to set up a Selenium 2.0 Java project is to use Maven. Maven will download the java bindings (the Selenium 2.0 java client library) and all its dependencies, and will create the project for you, using a maven pom.xml (project configuration) file. Once done, you can import the maven project into your preferred IDE, IntelliJ IDEA or Eclipse.

Advantages of Selenium

i) It is an Open source Software.
ii) It supports various Operating environments (Windows, Linux, Mac etc…)
iii) It supports various browsers (IE, Mozilla Firefox, Chrome, safari, Opera etc…)
iv) It supports various programming environments (Java, Perl, Python, Ruby and PHP)
v) It supports parallel Test execution.
vi) It uses less Hardware resources.

Disadvantages of Selenium

i) It supports Web based Applications only.
ii) No reliable support from anybody.
iii) No centralized maintenance of Elements/objects
iv) Difficult to setup environment.
v) Difficult to use.
vi) Limited support for Image based testing.
vii) New features may not work properly.
viii) No other tool integration for test management & No built in Reporting facility

- **Test Plan (e.g. given)**

INTRODUCTION

The Test Plan is designed to prescribe the scope, approach, resources, and schedule of all testing activities of the project Guru99 Bank.

The plan identify the items to be tested, the features to be tested, the types of testing to be performed, the personnel responsible for testing, the resources and schedule required to complete testing, and the risks associated with the plan.

1 TEST STRATEGY

1.1 Scope of Testing

1.1.1 Feature to be tested

All the feature of websiteGuru99 Bank which were defined in software requirement are need to bested

Module Name	Applicable Roles	Description
Balance Enquiry	Manager Customer	**Customer:** A customer can have multiple bank accounts. He can view balance of his accounts only **Manager:** A manager can view balance of all the customers who come under his supervision
Fund Transfer	Manager Customer	**Customer:** A customer can have transfer funds from his "own" account to any destination account. **Manager:** A manager can transfer funds from any source bank account to destination account
Mini Statement	Manager Customer	A Mini statement will show last 5 transactions of an account **Customer:** A customer can see mini-statement of only his "own" accounts **Manager:** A manager can see mini-statement of any account
Customized Statement	Manager Customer	A customized statement allows you to filter and display transactions in an account based on date, transaction value **Customer:** A customer can see Customized- statement of only his "own" accounts **Manager:** A manager can see Customized -statement of any account
Change Password	Manager Customer	**Customer:** A customer can change password of only his account. **Manager:** A manager can change password of only his account. He cannot change passwords of his customers
New Customer	Manager	**Manager:** A manager can add a new customer.
	Manager	**Manager:** A manager can edit details like address, email, telephone of a customer.
New Account	Manager	Currently system provides 2 types of accounts • Saving • Current A customer can have multiple saving accounts (one in his name, other in a joint name etc).

		He can have multiple current accounts for different companies he owns. Or he can have a multiple current and saving accounts. **Manager:** A manager can add a new account for an existing customer.
Edit Account	Manager	**Manager:** A manager can add a edit account details for an existing account
Delete Account	Manager	**Manager:** A manager can add a delete an account for a customer.
Delete Customer	Manager	A customer can be deleted only if he/she has no active current or saving accounts **Manager:** A manager can delete a customer.
Deposit	Manager	**Manager:** A manager can deposit money into any account. Usually done when cash is deposited at a bank branch.
Withdrawal	Manager	**Manager:** A manager can withdraw money from any account. Usually done when cash is withdrawn at a bank branch.

1.1.2 Feature not to be tested

These feature are not be tested because they are not included in the software requirement

- User Interfaces
- Hardware Interfaces
- Software Interfaces
- Database logical
- Communications Interfaces
- Website Security and Performance

1.2 Test Type

In the project Guru99 Bank, there're 3 types of testing should be conducted.

- Integration Testing (Individual software modules are combined and tested as a group)
- System Testing: Conducted on a complete, integrated system to evaluate the system's compliance with its specified requirements
- API testing: Test all the APIs create for the software under tested

1.3 Risk and Issues

Risk	Mitigation
Team member lack the required skills for website testing.	Plan **training course** to skill up your members
The project schedule is too tight; it's hard to complete this project on time	Set **Test Priority** for each of the test activity.
Test Manager has poor management skill	Plan leadership training for manager

A lack of cooperation negatively affects your employees' productivity	**Encourage** each team member in his task, **and inspire** them to greater efforts.
Wrong budget estimate and cost overruns	Establish the scope before beginning work, pay a lot of attention to project planning and constantly track and measure the progress

1.4 Test Logistics

1.4.1 Who will test?

The project should use outsource members as the tester to save the project cost

1.4.2 When will test occur?

The tester will start the test execution when all the following inputs are ready

- Software is available for testing
- Test Specification is created
- Test Environment is built
- Enough human resource for testing

2 TEST OBJECTIVE

The test objectives are to verify the Functionality of website Guru99 Bank, the project should focus on testing the banking operation such as Account Management, Withdrawal, and Balance…etc. to guarantee all these operation can work normally in real business environment.

3 TEST CRITERIA

3.1 Suspension Criteria

If the team members report that there are 40% of test cases failed, suspend testing until the development team fixes all the failed cases.

3.2 Exit Criteria

Specifies the criteria that denote a successful completion of a test phase

- Run rate is mandatory to be 100% unless a clear reason is given.
- Pass rate is 80%, achieving the pass rate is mandatory.

4 RESOURCE PLANNING

4.1 System Resource

No.	Resources	Descriptions
1.	Server	Need a Database server which install MySQL server Web server which install Apache Server

2.	Test tool	Develop a Test tool which can auto generate the test result to the pre-defined form and automated test execution
3.	Network	Setup a LAN Gigabit and 1 internet line with the speed at least 5 Mb/s
4.	Computer	At least 4 computer run Windows 7, Ram 2GB, CPU 3.4GHZ

4.2 Human Resource

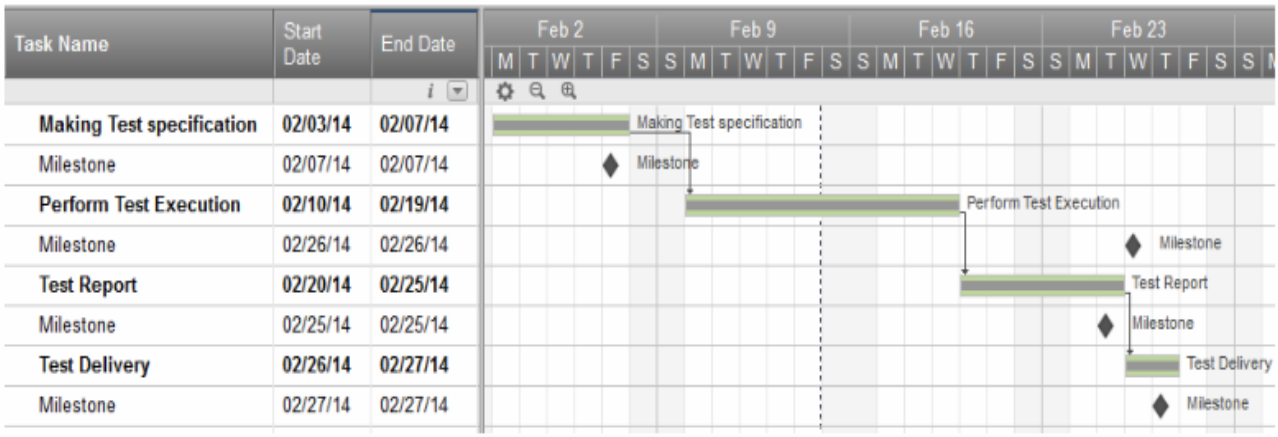
No.	Member	Tasks
1.	Test Manager	Manage the whole project Define project directions Acquire appropriate resources
2.	Test	Identifying and describing appropriate test techniques/tools/automation architecture Verify and assess the Test Approach Execute the tests, Log results, Report the defects. Outsourced members
3.	Developer in Test	Implement the test cases, test program, test suite etc.
4.	Test Administrator	Builds up and ensures test environment and assets are managed and maintained Support Tester to use the test environment for test execution
5.	SQA members	Take in charge of quality assurance Check to confirm whether the testing process is meeting specified requirements

5 SCHEDULE & ESTIMATION

5.1 All project task and estimation

Task	Members	Estimate effort
Create the test specification	Test Designer	170 man-hour
Perform Test Execution	Tester, Test Administrator	80 man-hour
Test Report	Tester	10 man-hour
Test Delivery		20 man-hour
Total		**280 man-hour**

Schedule to complete these task



TEST DELIVERABLES

Test deliverables are provided as below

7.1 Before testing phase

- Test plans document.
- Test cases documents
- Test Design specifications.

7.2 During the testing

- Test Tool
- Simulators.
- Test Data
- Test Trace-ability Matrix
- Error logs and execution logs.

7.3 After the testing cycles is over

- Test Results/reports
- Defect Report
- Installation/ Test procedures guidelines
- Release note

- Test Report for Test Cases inclusive of Test Procedures for identified Test Scenarios.

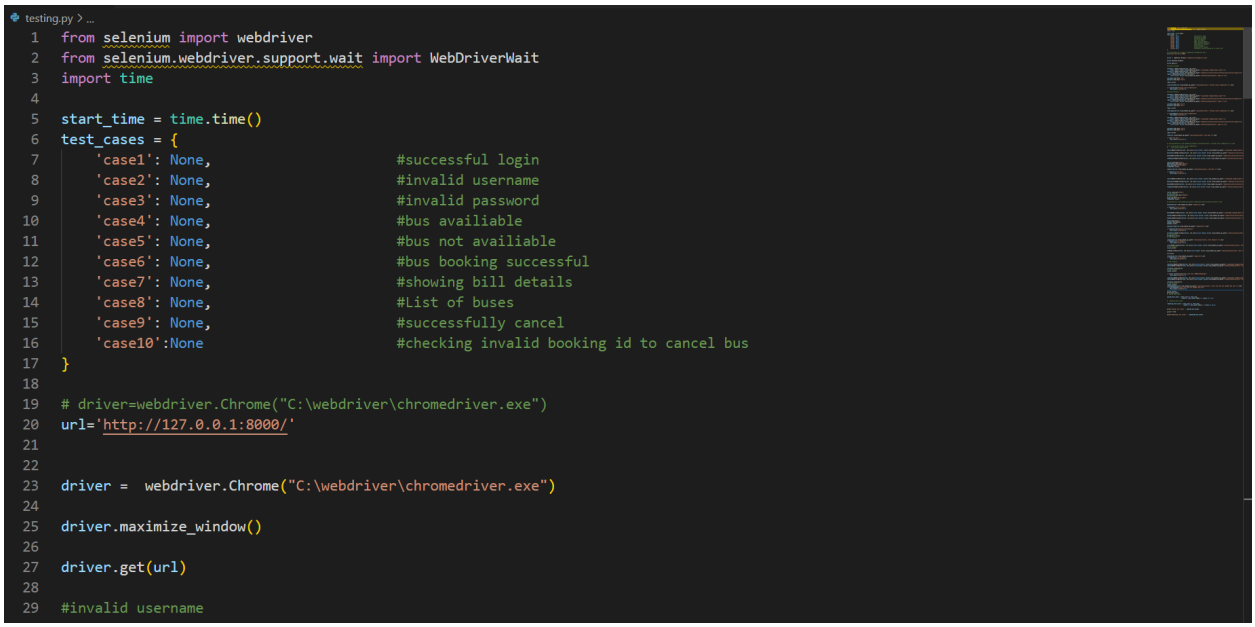
Sr. No.	Test Objective	Steps	Expected Result	Actual Result	Test Status
TC01	Check User Login	Enter Valid User name and password	Successful login	Successful Login	Pass
TC02	Check User Login	Enter Invalid User name	Message to be displayed "Provide valid credentials"	Message Displayed	Pass
TC0	Check User	Enter Invalid	Message to	Message	Pass

3	Login	Password	be displayed “Provide valid credentials ”	Displayed	
TC04	Check Bus availability.	Enter Bus details.	Show list of available bus	List Displayed	Pass
TC05	Check the message if bus is not available.	Enter Bus details.	Message to be displayed “Sorry no buses available”	Message Displayed	Pass
TC06	Check bus booking status	Enter Bus id and no. of seats.	Redirect to Booking Confirmation Page.	Redirected	Pass
TC07	Check the bill details with ticket details	Click on bill details	Bill details window pop up	Window displayed	Pass

TC08	Check all the list of buses	Press ok button.	List of buses should be displayed.	List displayed.	Pass
TC09	Check seat cancel functionality	Enter valid booking id.	Current booking should be cancelled.	Booking cancelled.	Pass

TC10	Check Invalid booking id to cancel the bus.	Enter invalid Booking id.	Message to be displayed “Sorry you have not booked that bus”	Message displayed.	Pass
Add minimum 30 test case					

- **Screenshot of Code and Output**
- **(2-3 screenshots of GUI)**



```python
from selenium import webdriver
from selenium.webdriver.support.wait import WebDriverWait
import time

start_time = time.time()
test_cases = {
    'case1': None,                  #successful login
    'case2': None,                  #invalid username
    'case3': None,                  #invalid password
    'case4': None,                  #bus availiable
    'case5': None,                  #bus not availiable
    'case6': None,                  #bus booking successful
    'case7': None,                  #showing bill details
    'case8': None,                  #List of buses
    'case9': None,                  #successfully cancel
    'case10':None                   #checking invalid booking id to cancel bus
}

# driver=webdriver.Chrome("C:\webdriver\chromedriver.exe")
url='http://127.0.0.1:8000/'


driver =  webdriver.Chrome("C:\webdriver\chromedriver.exe")

driver.maximize_window()

driver.get(url)

#invalid username
```

```
username = WebDriverWait(driver, 30).until(
    lambda driver: driver.find_element_by_xpath("//input[@id='example-email-input']"))
password = WebDriverWait(driver, 30).until(
    lambda driver: driver.find_element_by_xpath("//body/div[1]/div[1]/div[1]/form[1]/div[2]/div[1]/input[1]"))
login = WebDriverWait(driver, 30).until(
    lambda driver: driver.find_element_by_xpath("//button[contains(text(),'Sign in')]"))

username.send_keys('rot')
password.send_keys('root')

login.click()

invalidcred=driver.find_element_by_xpath("//p[contains(text(),'Provide valid credentials')]").text

if invalidcred=="Provide valid credentials":
    test_cases["case2"]=True

#invalid Password

username = WebDriverWait(driver, 30).until(
    lambda driver: driver.find_element_by_xpath("//input[@id='example-email-input']"))
password = WebDriverWait(driver, 30).until(
    lambda driver: driver.find_element_by_xpath("//body/div[1]/div[1]/div[1]/form[1]/div[2]/div[1]/input[1]"))
login = WebDriverWait(driver, 30).until(
    lambda driver: driver.find_element_by_xpath("//button[contains(text(),'Sign in')]"))

username.send_keys('root')
password.send_keys('oot')
```

```
if invalidpass=="Provide valid credentials":
    test_cases["case3"]=True

username = WebDriverWait(driver, 30).until(
    lambda driver: driver.find_element_by_xpath("//input[@id='example-email-input']"))
password = WebDriverWait(driver, 30).until(
    lambda driver: driver.find_element_by_xpath("//body/div[1]/div[1]/div[1]/form[1]/div[2]/div[1]/input[1]"))
login = WebDriverWait(driver, 30).until(
    lambda driver: driver.find_element_by_xpath("//button[contains(text(),'Sign in')]"))


username.send_keys('root')
password.send_keys('root')

login.click()

fb=driver.find_element_by_xpath("//h2[contains(text(),'Find bus')]").text

if fb=="Find bus":
    test_cases["case1"]=True


# invalidcred=driver.find_element_by_xpath("//p[contains(text(),'Provide valid credentials')]").text

# if invalidcred=="Provid valid credentials":
#     test_cases["case2"]=True

source=WebDriverWait(driver, 30).until(lambda driver: driver.find_element_by_xpath("//input[@id='example-email-input']"))
```

```python
ok.click()

lstbuses=driver.find_element_by_xpath("//body//h2").text
if lstbuses=="List of buses":
    test_cases["case8"]=True

# time.sleep(3)

cancelbus=WebDriverWait(driver, 30).until(lambda driver: driver.find_element_by_xpath("//input[@id='example-email-input']"))
cancel=WebDriverWait(driver, 30).until(lambda driver: driver.find_element_by_xpath("//button[contains(text(),'Cancel bus')]"))

cancelbus.send_keys(37)
# time.sleep(5)
cancel.click()

if driver.current_url=="http://127.0.0.1:8000/seebookings":
    test_cases["case9"]=True

cancelbus=WebDriverWait(driver, 30).until(lambda driver: driver.find_element_by_xpath("//input[@id='example-email-input']"))
cancel=WebDriverWait(driver, 30).until(lambda driver: driver.find_element_by_xpath("//button[contains(text(),'Cancel bus')]"))

cancelbus.send_keys(2)
# time.sleep(5)
cancel.click()
invalidbookid=driver.find_element_by_xpath("//h1[contains(text(),'Sorry You have not booked that bus')]").text
if invalidbookid=="Sorry You have not booked that bus":
    test_cases["case10"]=True
```

```python
cancelbus=WebDriverWait(driver, 30).until(lambda driver: driver.find_element_by_xpath("//input[@id='example-email-input']"))
cancel=WebDriverWait(driver, 30).until(lambda driver: driver.find_element_by_xpath("//button[contains(text(),'Cancel bus')]"))

cancelbus.send_keys(2)
# time.sleep(5)
cancel.click()
invalidbookid=driver.find_element_by_xpath("//h1[contains(text(),'Sorry You have not booked that bus')]").text
if invalidbookid=="Sorry You have not booked that bus":
    test_cases["case10"]=True

driver.close()
print(test_cases)
#  passed test cases

passed_test_cases = [test_case for test_case,
                     status in test_cases.items() if status == True]

#  remaining test cases

remaining_test_cases = [test_case for test_case,
                        status in test_cases.items() if status == None]


print('Passed test cases: ', passed_test_cases)

print("*"*50)

print('Remaining test cases: ', remaining_test_cases)
```

```
DevTools listening on ws://127.0.0.1:51724/devtools/browser/e3650d17-b62c-4ffb-9aff-8ecb7af1e642
{'case1': True, 'case2': True, 'case3': True, 'case4': True, 'case5': True, 'case6': True, 'case7': True, 'case8': True, 'case9': True, 'case10': True}
Passed test cases:  ['case1', 'case2', 'case3', 'case4', 'case5', 'case6', 'case7', 'case8', 'case9', 'case10']
**************************************************
Remaining test cases:  []
PS D:\stqa_project\Bus-reservation-system-in-python-django-2.1>
```